

ORF 524 — Midterm Exam

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October 23, 2024

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Notes:

- Duration of examination: 180 minutes.
An additional 15 minutes are given to upload your solutions to Gradescope.
- All parts are equally weighted (5 points).
Please provide as much detail as possible in your answers.
Answers without proper justification will not receive full credit.
- Please start each question on a separate page.
A single PDF file must be submitted to Gradescope.
- Only one page of personal notes are allowed during the examination.
This is otherwise a closed-book and individual examination.

1 MSE and Point Estimation (50 points)

Assume $X \sim \mathbb{P}_\theta$ with $\theta \in \Theta$. Let $\hat{\theta}_n = \hat{\theta}_n(X)$ be a square-integrable estimator of θ , not necessarily unbiased. Suppose that $T(X)$ is a sufficient statistic for θ , and define:

$$\tilde{\theta}_n = \tilde{\theta}_n(X) = \mathbb{E}_\theta[\hat{\theta}_n | T(X)].$$

1. Show that $\tilde{\theta}_n$ is an estimator of θ .
2. Show that $\tilde{\theta}_n$ is unbiased (for θ) if and only if $\hat{\theta}_n$ is unbiased (for θ).
3. Show that $\mathbb{V}_\theta[\tilde{\theta}_n] \leq \mathbb{V}_\theta[\hat{\theta}_n]$.
4. Show that $\mathbb{E}_\theta[(\tilde{\theta}_n - \theta)^2] \leq \mathbb{E}_\theta[(\hat{\theta}_n - \theta)^2]$.
5. Determine whether the following statement is true or false:

$$\text{MSE}_\theta[\tilde{\theta}_n] \leq \text{MSE}_\theta[\hat{\theta}_n] \iff \mathbb{E}_\theta[\hat{\theta}_n] = \theta \quad \forall \theta \in \Theta.$$

where $\text{MSE}_\theta[Y] = \mathbb{E}_\theta[(Y - \theta)^2]$ for any square-integrable random variable Y .

Discuss the connections between this question and the Rao-Blackwell Theorem.

Let (U_1, U_2) be a random sample from the distribution $\text{Uniform}(0, \theta)$, where $\Theta = (0, \infty)$. Recall that $T(X) = \max(U_1, U_2)$ is a complete minimal sufficient statistics for θ .

6. Show that

$$F_{U_1 | \max(U_1, U_2)}(u|v) = \begin{cases} 0 & \text{if } u \leq 0 \\ \frac{u}{2v} & \text{if } 0 < u < v < \theta \\ 1 & \text{if } v \leq u \end{cases}$$

7. Show that $\mathbb{E}_\theta[U_1 | \max(U_1, U_2)] = \frac{3}{4} \max(U_1, U_2)$.
8. Show that $\hat{\theta}_n = U_1 + U_2$ is an unbiased estimator of θ .
9. Show that $\tilde{\theta}_n = \frac{3}{2} \max(U_1, U_2)$, and that it is also an unbiased estimator of θ .
10. Let $\check{\theta}_c = c \max(U_1, U_2)$, for $c \in \mathbb{R}$. Show that

$$\frac{\text{MSE}_\theta[\check{\theta}]}{\text{MSE}_\theta[\hat{\theta}]} = \frac{3}{4} < 1, \quad \text{and} \quad \frac{\text{MSE}_\theta[\check{\theta}_c]}{\text{MSE}_\theta[\hat{\theta}]} = \frac{8}{9} < 1,$$

for appropriate choice of c .

Explain in which sense $\check{\theta}$ is an optimal point estimator of θ , and when $\check{\theta}_c$ may be regarded as a better point estimator than $\tilde{\theta}$ (for appropriate choice of c).

2 MLE Methods: Poisson Distribution (50 points)

Let $(X_i : 1 \leq i \leq n)$ be an i.i.d. random sample from $X \sim \mathbb{P}_\theta = \text{Poisson}(\theta)$, with $\theta \in (0, \infty)$.

1. Find the Maximum Likelihood (ML) estimator of θ , denoted by $\hat{\theta}_{\text{ML}}$.
2. Show that $\sqrt{n}(\hat{\theta}_{\text{ML}} - \theta) \rightsquigarrow \text{N}(0, \theta)$.
3. Is $\hat{\theta}_{\text{ML}}$ asymptotically efficient for θ ?
4. Propose a feasible, asymptotically valid $100(1 - \alpha)\%$ confidence interval estimator for θ , denoted by $\text{CI}_{1-\alpha}(\theta)$, for $\alpha \in [0, 1]$.

Consider now the parameter $\beta = \beta(\theta) = \mathbb{P}_\theta[X_i = 0]$.

5. Find the ML estimator of β , denoted by $\hat{\beta}_{\text{ML}}$.
6. Show that $\sqrt{n}(\hat{\beta}_{\text{ML}} - \beta) \rightsquigarrow \text{N}(0, \theta \exp(-2\theta))$.
7. Is $\hat{\beta}_{\text{ML}}$ asymptotically efficient for β ?
8. Propose a feasible, asymptotically valid $100(1 - \alpha)\%$ confidence interval estimator for β , denoted by $\text{CI}_{1-\alpha}(\beta)$.

Consider the following one-sided hypothesis testing problem:

$$H_0 : \beta \geq \beta_0 \quad \text{vs.} \quad H_1 : \beta < \beta_0, \quad \text{for some fixed } \beta_0 \in (0, 1).$$

9. Propose a feasible, asymptotically α -level valid hypothesis test.
10. Show that the proposed hypothesis test is consistent.

Table 1.1. Discrete Distributions on $\mathcal{R}$

Uniform	p.d.f.	$1/m, x = a_1, \dots, a_m$
	m.g.f.	$\sum_{j=1}^m e^{a_j t}/m, t \in \mathcal{R}$
$DU(a_1, \dots, a_m)$	Expectation	$\sum_{j=1}^m a_j/m$
	Variance	$\sum_{j=1}^m (a_j - \bar{a})^2/m, \bar{a} = \sum_{j=1}^m a_j/m$
	Parameter	$a_i \in \mathcal{R}, m = 1, 2, \dots$
Binomial	p.d.f.	$\binom{n}{x} p^x (1-p)^{n-x}, x = 0, 1, \dots, n$
	m.g.f.	$(pe^t + 1 - p)^n, t \in \mathcal{R}$
$Bi(p, n)$	Expectation	np
	Variance	$np(1-p)$
	Parameter	$p \in [0, 1], n = 1, 2, \dots$
Poisson	p.d.f.	$\theta^x e^{-\theta}/x!, x = 0, 1, 2, \dots$
	m.g.f.	$e^{\theta(e^t - 1)}, t \in \mathcal{R}$
$P(\theta)$	Expectation	θ
	Variance	θ
	Parameter	$\theta > 0$
Geometric	p.d.f.	$(1-p)^{x-1}p, x = 1, 2, \dots$
	m.g.f.	$pe^t/[1 - (1-p)e^t], t < -\log(1-p)$
$G(p)$	Expectation	$1/p$
	Variance	$(1-p)/p^2$
	Parameter	$p \in [0, 1]$
Hyper-geometric	p.d.f.	$\binom{n}{x} \binom{m}{r-x} / \binom{N}{r}$
		$x = 0, 1, \dots, \min\{r, n\}, r - x \leq m$
$HG(r, n, m)$	m.g.f.	No explicit form
	Expectation	rn/N
	Variance	$rn m(N-r)/[N^2(N-1)]$
	Parameter	$r, n, m = 1, 2, \dots, N = n + m$
Negative binomial	p.d.f.	$\binom{x-1}{r-1} p^r (1-p)^{x-r}, x = r, r+1, \dots$
	m.g.f.	$p^r e^{rt}/[1 - (1-p)e^t]^r, t < -\log(1-p)$
$NB(p, r)$	Expectation	r/p
	Variance	$r(1-p)/p^2$
	Parameter	$p \in [0, 1], r = 1, 2, \dots$
Log-distribution	p.d.f.	$-(\log p)^{-1} x^{-1} (1-p)^x, x = 1, 2, \dots$
	m.g.f.	$\log[1 - (1-p)e^t]/\log p, t \in \mathcal{R}$
$L(p)$	Expectation	$-(1-p)/(p \log p)$
	Variance	$-(1-p)[1 + (1-p)/\log p]/(p^2 \log p)$
	Parameter	$p \in (0, 1)$

All p.d.f.'s are w.r.t. counting measure.

Table 1.2. Distributions on $\mathcal{R}$ with Lebesgue p.d.f.'s

Uniform	p.d.f.	$(b-a)^{-1}I_{(a,b)}(x)$
	m.g.f.	$(e^{bt} - e^{at})/[(b-a)t], \quad t \in \mathcal{R}$
$U(a, b)$	Expectation	$(a+b)/2$
	Variance	$(b-a)^2/12$
	Parameter	$a, b \in \mathcal{R}, \quad a < b$
Normal	p.d.f.	$\frac{1}{\sqrt{2\pi}\sigma} e^{-(x-\mu)^2/2\sigma^2}$
	m.g.f.	$e^{\mu t + \sigma^2 t^2/2}, \quad t \in \mathcal{R}$
$N(\mu, \sigma^2)$	Expectation	μ
	Variance	σ^2
	Parameter	$\mu \in \mathcal{R}, \quad \sigma > 0$
Exponential	p.d.f.	$\theta^{-1} e^{-(x-a)/\theta} I_{(a, \infty)}(x)$
	m.g.f.	$e^{at} (1 - \theta t)^{-1}, \quad t < \theta^{-1}$
$E(a, \theta)$	Expectation	$\theta + a$
	Variance	θ^2
	Parameter	$\theta > 0, \quad a \in \mathcal{R}$
Chi-square	p.d.f.	$\frac{1}{\Gamma(k/2)2^{k/2}} x^{k/2-1} e^{-x/2} I_{(0, \infty)}(x)$
	m.g.f.	$(1 - 2t)^{-k/2}, \quad t < 1/2$
χ_k^2	Expectation	k
	Variance	$2k$
	Parameter	$k = 1, 2, \dots$
Gamma	p.d.f.	$\frac{1}{\Gamma(\alpha)\gamma^\alpha} x^{\alpha-1} e^{-x/\gamma} I_{(0, \infty)}(x)$
	m.g.f.	$(1 - \gamma t)^{-\alpha}, \quad t < \gamma^{-1}$
$\Gamma(\alpha, \gamma)$	Expectation	$\alpha\gamma$
	Variance	$\alpha\gamma^2$
	Parameter	$\gamma > 0, \quad \alpha > 0$
Beta	p.d.f.	$\frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} x^{\alpha-1} (1-x)^{\beta-1} I_{(0,1)}(x)$
	m.g.f.	No explicit form
$B(\alpha, \beta)$	Expectation	$\alpha/(\alpha + \beta)$
	Variance	$\alpha\beta/[(\alpha + \beta + 1)(\alpha + \beta)^2]$
	Parameter	$\alpha > 0, \quad \beta > 0$
Cauchy	p.d.f.	$\frac{1}{\pi\sigma} \left[1 + \left(\frac{x-\mu}{\sigma} \right)^2 \right]^{-1}$
	ch.f.	$e^{\sqrt{-1}\mu t - \sigma t }$
$C(\mu, \sigma)$	Expectation	Does not exist
	Variance	Does not exist
	Parameter	$\mu \in \mathcal{R}, \quad \sigma > 0$

Table 1.2. (continued)

t-distribution	p.d.f.	$\frac{\Gamma[(n+1)/2]}{\sqrt{n\pi}\Gamma(n/2)} \left(1 + \frac{x^2}{n}\right)^{-(n+1)/2}$
	ch.f.	No explicit form
t_n	Expectation	0, $(n > 1)$
	Variance	$n/(n-2)$, $(n > 2)$
	Parameter	$n = 1, 2, \dots$
F-distribution	p.d.f.	$\frac{n^{n/2}m^{m/2}\Gamma[(n+m)/2]x^{n/2-1}}{\Gamma(n/2)\Gamma(m/2)(m+nx)^{(n+m)/2}}I_{(0,\infty)}(x)$
	ch.f.	No explicit form
$F_{n,m}$	Expectation	$m/(m-2)$, $(m > 2)$
	Variance	$2m^2(n+m-2)/[n(m-2)^2(m-4)]$, $(m > 4)$
	Parameter	$n = 1, 2, \dots$, $m = 1, 2, \dots$
Log-normal	p.d.f.	$\frac{1}{\sqrt{2\pi}\sigma}x^{-1}e^{-(\log x - \mu)^2/2\sigma^2}I_{(0,\infty)}(x)$
	ch.f.	No explicit form
$LN(\mu, \sigma^2)$	Expectation	$e^{\mu+\sigma^2/2}$
	Variance	$e^{2\mu+\sigma^2}(e^{\sigma^2} - 1)$
	Parameter	$\mu \in \mathcal{R}$, $\sigma > 0$
Weibull	p.d.f.	$\frac{\alpha}{\theta}x^{\alpha-1}e^{-x^\alpha/\theta}I_{(0,\infty)}(x)$
	ch.f.	No explicit form
$W(\alpha, \theta)$	Expectation	$\theta^{1/\alpha}\Gamma(\alpha^{-1} + 1)$
	Variance	$\theta^{2/\alpha} \left\{ \Gamma(2\alpha^{-1} + 1) - [\Gamma(\alpha^{-1} + 1)]^2 \right\}$
	Parameter	$\theta > 0$, $\alpha > 0$
Double Exponential	p.d.f.	$\frac{1}{2\theta}e^{- x-\mu /\theta}$
	m.g.f.	$e^{\mu t}/(1 - \theta^2 t^2)$, $ t < \theta^{-1}$
$DE(\mu, \theta)$	Expectation	μ
	Variance	$2\theta^2$
	Parameter	$\mu \in \mathcal{R}$, $\theta > 0$
Pareto	p.d.f.	$\theta a^\theta x^{-(\theta+1)}I_{(a,\infty)}(x)$
	ch.f.	No explicit form
$Pa(a, \theta)$	Expectation	$\theta a/(\theta - 1)$, $(\theta > 1)$
	Variance	$\theta a^2/[(\theta - 1)^2(\theta - 2)]$, $(\theta > 2)$
	Parameter	$\theta > 0$, $a > 0$
Logistic	p.d.f.	$\sigma^{-1}e^{-(x-\mu)/\sigma}/[1 + e^{-(x-\mu)/\sigma}]^2$
	m.g.f.	$e^{\mu t}\Gamma(1 + \sigma t)\Gamma(1 - \sigma t)$, $ t < \sigma^{-1}$
$LG(\mu, \sigma)$	Expectation	μ
	Variance	$\sigma^2\pi^2/3$
	Parameter	$\mu \in \mathcal{R}$, $\sigma > 0$